MIL-C-29104A(SA)
18 August 1976
SUPERSEDING
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22 June 1971

MILITARY SPECIFICATION

CLOTH, TROPICAL, POLYESTER AND RAYON

This specification is approved for use by the Naval Supply Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a polyester and rayon spun yarn tropical cloth in one type only.

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-285 - Thread, Polyester
PPP-P-1133 - Packaging and Packing of Synthetic Fiber Fabrics

STANDARDS

FEDERAL

FED-STD-4 - Glossary of Fabric Imperfections
FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 8305

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

RULES AND REGULATIONS UNDER THE
TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Application for copies should be addressed to Federal Trade Commission, Washington, D.C. 20580.)

3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade, uniformity of shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article approval. The requirements for first article shall be as specified by the procuring activity (see 6.2).

3.3 Material.

3.3.1 Fiber blend.

3.3.1.1 Polyester fiber. The fibers shall be polyethylene glycol terephthalate dull or semi-dull when tested as specified in 4.5.

3.3.1.2 Rayon fiber. The fiber shall be rayon staple when tested as specified in 4.5. The use of rayon waste of any kind is prohibited.

* 3.3.2 Blended yarn. The polyester fibers and rayon fibers shall be blended and evenly spun into a 2-ply yarn for both warp and filling. The yarn shall be a blend of 65 percent polyester and 35 percent rayon. A tolerance of plus or minus 3.0 percent on fiber content is permissable when tested as specified in 4.5.

* 3.4 Color. The color of the finished cloth shall be white 3053 conforming to the standard sample when tested as specified in 4.5. All cloth within any one contract shall be processed for whiteness in the same manner. The color shall be obtained with a chemical bleach, preferably with hydrogen peroxide or peroxygen chemicals. The chemical bleach may be supplemented with a fluorescent whiting agent.

3.4.1 Discoloration. The cloth shall show no more discoloration to light, perspiration and wet drycleaning than that of the standard sample when tested as specified in 4.5.

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3.5 Physical requirements. The finished cloth shall conform to the requirements specified in Table I when tested as specified in 4.5.

Table I - Physical requirements

Weight Oz/sq yd	Yarns per inch (minimum) Warp	Filling	Breaking strength (minimum) (pounds) Warp	Filling	Stiffness (inch pound) Minimum Warp	Filling	Weave
5.5 (±.5)	58	48	125	110	.0007 .0006	.0003 1/ .0003 2/	Plain

1/ Initial
2/ After 3 launderings

3.5.1 Width. The minimum width (inclusive of the selvages) of the finished cloth shall be as specified (see 6.2).

3.6 Nonfibrous materials. The total nonfibrous material content of resin finished cloth shall not exceed 12 percent when tested as specified in 4.5.

3.7 Finish. The cloth shall be scoured, heat set, bleached and finished with a resin to provide a finish and hand equal in characteristics to that of the standard sample and to meet the requirements specified herein. The resin finish shall be of a non-chlorine retentive type and the material shall show no more discoloration and no more than 5.0 percent loss of its original breaking strength when tested as specified in 4.5. The resin treated cloth shall be processed at such time and temperature cycle that will insure adequate curing of the resin, and should be washed thoroughly to remove all traces of uncured resin. The use of a silicone impregnate to give desired softness is prohibited.

3.8 pH. The pH value of the water extract of the finished cloth shall be not less than 5.0 nor more than 8.5 when tested as specified in 4.5.

3.9 Shrinkage. The cloth shall be preshrunk so that it shall not shrink more than 2.0 percent in the warp or filling when tested as specified in 4.5. The preshrinking process shall not be identified by name or trademark either on the cloth or on the ticket or package.

3.10 Seam efficiency. The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.5.

3.11 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up on full width rolls as specified in PPP-P-1133.

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3.11.1 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.12 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.13 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (83.6 m^2) when calculated as specified in Section 4 shall not exceed the established maximum point value.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 First article inspection. When required, the first article of the finished cloth shall be visually inspected for shade, finish and appearance and shall be tested for the chemical and physical properties in accordance with the applicable methods specified in 4.5.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except when otherwise indicated.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.2 Examination of the end item. Defects found during examination shall be classified in accordance with 4.3.2.1 thru 4.3.2.4.

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* 4.3.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards (91.4 m), the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards (91.4 m), only 100 yards (91.4 m) shall be examined. Defects as defined in Section III of FED-STD-4 which are clearly noticeable at normal inspection distance (3 feet)(.91 m) shall be scored and assigned demerit points as listed in 4.3.2.1.1 except that only those slubs and knots which exceed the limits shown on Sears Fabric Defect Scale (see 6.4), "D" for slubs and "C" for knots, shall be scored. No contiguous linear yard or increment of one yard (.91 m) on the measuring device of the inspection machine from any one roll within the sample, shall be penalized more than 4 points. The sample unit for this examination shall be one linear yard. The sample size shall be 20 rolls selected from 20 containers. For determination of acceptability on the average quality of the lot, the lot shall be unacceptable if the points per 100 square yards (83.6 m^2) exceeds the following:

25.0 points

For determination of acceptability on the individual roll quality basis, the lot shall be unacceptable if two or more individual rolls exceed the following:

30.0 points

If no individual roll exceeds the point level, the lot shall be accepted. If one roll exceeds the point level, a second sample of 20 rolls shall be selected for individual roll quality examination only. If no roll in the second sample exceeds the point value, the lot shall be accepted. If one or more rolls exceeds the value, the lot shall be rejected. Point computation for the lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.3.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defect 3 inches (7.62 cm) or less in any dimension	- one point
For defects exceeding 3 inches (7.62 cm), but not exceeding 6 inches (15.24 cm) in any dimension	- two points
For defects exceeding 6 inches (15.24 cm), but not exceeding 9 inches (22.86 cm) in any dimension	- three points
For defects exceeding 9 inches (22.86 cm) in any dimension	- four points

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NOTE: The following defects when present shall be scored four points for each yard in which they occur.

Objectionable odor.
Width of cloth less than specified.
Excessive neppiness.
Characteristics of finish, hand and shade not equal to the standard sample.
Baggy, ridgy or wavy cloth.

4.3.2.2 Examination for length.

4.3.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.3.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of lengths marked on the ticket. The rolls examined shall be those selected for the examination of individual rolls.

4.3.2.3 Examination for face identification, identification of preshrinkage process and compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination each roll in the sample shall be examined for these defects. The lot shall be unacceptable if two or more rolls in the sample have face identification missing from either or both ends, contain identification of the preshrinkage process by name or trade mark on the cloth or ticket, or are not labelled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing and marking complies with Section 5 requirements of this specification.

4.5 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table II, shall be followed. The physical and chemical values specified in Section 3 except when otherwise indicated apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be 4-1/2 continuous yards (3.68 m) full width of the finished cloth. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of 1 yard (.91 m). The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

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Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

Table II - Test methods

Characteristic	Requirement paragraph	Test method
Polyester fiber identification	3.3.1.1	1/
Rayon fiber identification	3.3.1.2	1/
Polyester fiber content	3.3.2	4.5.1 and 2100
Rayon fiber content	3.3.2	4.5.1 and 2100
Yarn ply	3.3.2	1/
Bleaching agents/Brighteners	3.4	1/
Discoloration to:		
Light	3.4.1	2/
Perspiration	3.4.1	5680
Wet drycleaning	3.4.1	5622
Stiffness, bending moment:		
Before laundering	3.5	5202
After 3 launderings	3.5	5552 and 5202 5/
Weight	3.5	5041
Yarns per inch:		
Warp	3.5	5050
Filling	3.5	5050
Breaking strength:		
Warp	3.5	5100
Filling	3.5	5100
Weave	3.5	Visual 3/
Nonfibrous material content	3.6	2610
Finish:		
Scoured	3.7	1/
Heat set	3.7	1/
Nonchlorine retentive resin	3.7	4.5.2 and 4.5.3
Chlorine degradation	3.7	4.5.2
Discoloration	3.7	4.5.3
pH	3.8	2811
Shrinkage in laundering	3.9	5552
Seam efficiency	3.10	5110 4/

1/ Unless otherwise specified a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

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2/ Method 5660 shall be used except that the exposure time shall be 20 standard fading hours and the exposed sample shall be allowed to lie in the dark at room temperature for a minimum of 48 hours before comparison. Any specimen showing discoloration less than or equal to that shown by the standard sample shall be considered satisfactory. One determination shall be made for each sample unit and the result reported as "pass" or "fail".

3/ One determination per sample unit and result reported as "pass" or "fail".

4/ The needle size shall be .040 inch (+ .001 inch). The needle thread shall be size B polyester yarn and the looper thread shall be size A polyester yarn in accordance with type I of V-T-285.

5/ The cotton laundering procedure shall be used.

4.5.1 Test for polyester fiber and rayon fiber content. The general procedures of Test Method 2100 shall be followed and the fiber content percentages shall be calculated as follows:

$$\frac{\text{Weight of dry residual fiber} \times 100}{\text{Weight of dry desized specimen}} = \text{percent polyester}$$

100 - percent polyester = percent rayon

4.5.1.1 Report. Two specimens shall be tested from each sample unit and the average percent polyester and the average percent rayon of the two specimens shall be reported to the nearest 0.1 percent.

* 4.5.2 Chlorine Degradation. A piece of cloth not less than 20 inches (50.8 cm) long and full width, shall be laundered in accordance with Method 5552 of FED-STD-191 and shall then be treated for one hour with 0.7 percent available chlorine solution with a temperature of 25° (+ 1°)C. The piece shall be rinsed in a sour solution of 0.014 percent acetic acid (0.005 ml. of 28 percent acetic acid per 100 ml. of water) for one minute at 27°C (+ 3°C), and washed thoroughly until a control piece (undyed, desized, cotton cloth) shows no trace of chlorine. The piece shall be air-dried and then conditioned under Standard Conditions as defined in FED-STD-191. Eight pairs of specimens, four warpwise and four fillingwise, with each set having segments of the same yarns, shall be taken for tests. Four warp specimens and four filling specimens shall be tested in accordance with Method 5100 of FED-STD-191. The remaining warp and filling specimens shall be oven dried at 105 to 110°C for two hours and tested in the same manner as the first eight specimens. The difference in the average breaking strength of the respective warp and filling samples shall be computed and the percentage of loss determined.

NOTE: It should be noted, that the cloth to be used as ballast, in Test Method 5552 shall be white in color.

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4.5.3 Discoloration test. A 4 by 4 inch (10.16 by 10.16 cm) piece of the chlorinated and conditioned sample of cloth from 4.5.2 with a piece of the unchlorinated cloth, shall be oven dried at a temperature of 105 to 110°C for two hours. Both pieces shall be removed from the oven and allowed to cool at room temperature. The chlorinated piece shall then be compared with the control cloth for discoloration. Any discoloration shall be a cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2 and 6.2.1).

5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 Levels A, B and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth is intended for use in the manufacture of a white summer dress uniform worn by Naval female personnel.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Requirements for first article approval (see 3.2).
c. Width required (see 3.5.1).
d. Minimum length if other than specified (see 3.11).,
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in PPP-P-1133, may be exercised in the preparation for delivery of this cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

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* 6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 733Q, "Sears Defect Replica Kit", Sears Tower, Chicago, Illinois 60684.

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-C-29104A(SA) - CLOTH, TROPICAL, POLYESTER AND RAYON

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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